

Ethanol Blending in Gasoline – El Salvador

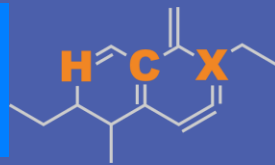
August 2025



**U.S. GRAINS &
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Ethanol Blending in Latin America

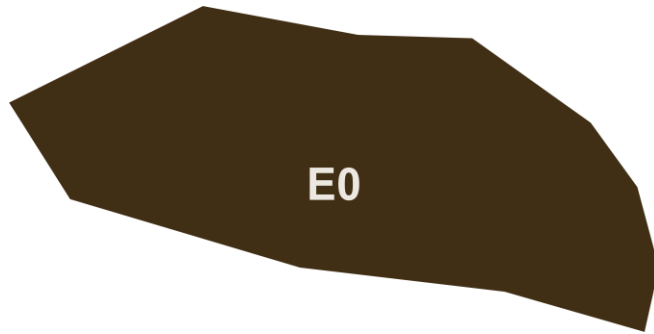
There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Latin America to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.

Sixteen countries with potential and additional use of ethanol were studied: 1) gasoline market profiles; 2) Optimization of gasoline blends with ethanol and 3) Environmental impact of gasolines blended with ethanol.



Ethanol Blending in Gasoline - El Salvador



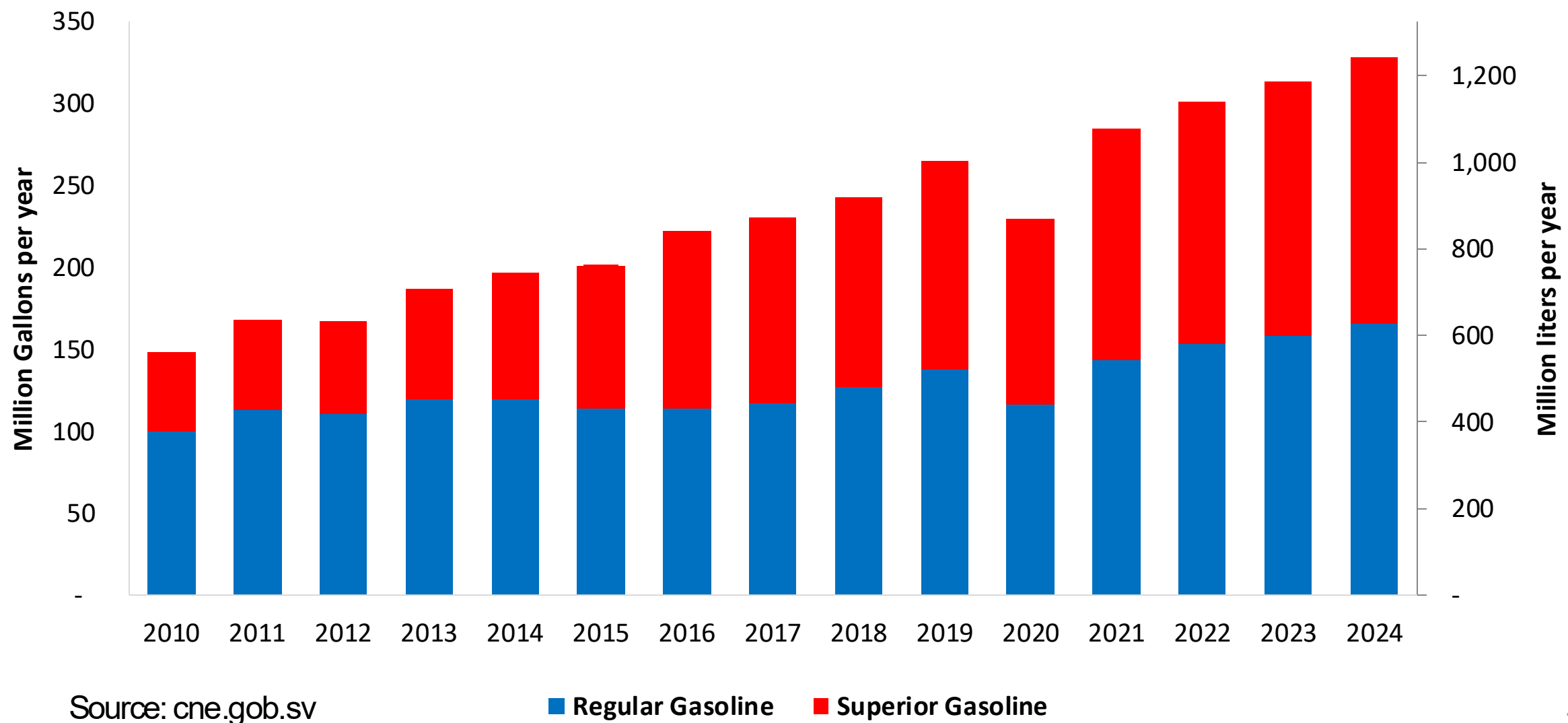
In 2024, gasoline demand was 328 million gallons (1,243 million liters). Regular gasoline RON 91 represented 51% of the volume consumed while Superior gasoline RON 95 reached 49%. Gasoline is supplied only by imports, distributed mainly by Chevron, Puma and Uno companies and imported mainly from the United States.

El Salvador has no national mandate for ethanol. It has the potential to produce ethanol from maize and sugar cane. In the last year there has been an interest for its use.

Sources: cne.gob.sv

Gasoline Demand in El Salvador

Gasoline Demand by Grade in El Salvador



Source: cne.gob.sv

The FARO90 logo is shown in a blue box, followed by a chemical structure diagram of a substituted cyclohexene ring. The structure features a cyclohexene ring with a methyl group at position 1, a vinyl group at position 2, and a substituent 'X' at position 3. The positions 1, 2, and 3 are labeled with orange letters H, C, and X respectively.

This stacked bar chart illustrates the annual consumption of Regular Gasoline and Superior Gasoline from 2010 to 2024. The left Y-axis measures consumption in Million Gallons per year (0 to 350), while the right Y-axis measures it in Million liters per year (0 to 1,200). Regular Gasoline is represented by blue bars, and Superior Gasoline is represented by red bars. The total consumption shows a general upward trend over the period, with a notable dip in 2020.

Year	Regular Gasoline (Million Gallons)	Superior Gasoline (Million Gallons)	Total (Million Gallons)
2010	98	48	146
2011	110	55	165
2012	107	57	164
2013	116	68	184
2014	116	78	194
2015	116	87	203
2016	112	105	217
2017	111	125	236
2018	127	115	242
2019	133	130	263
2020	104	97	201
2021	145	142	287
2022	149	153	302
2023	153	154	307
2024	161	163	324

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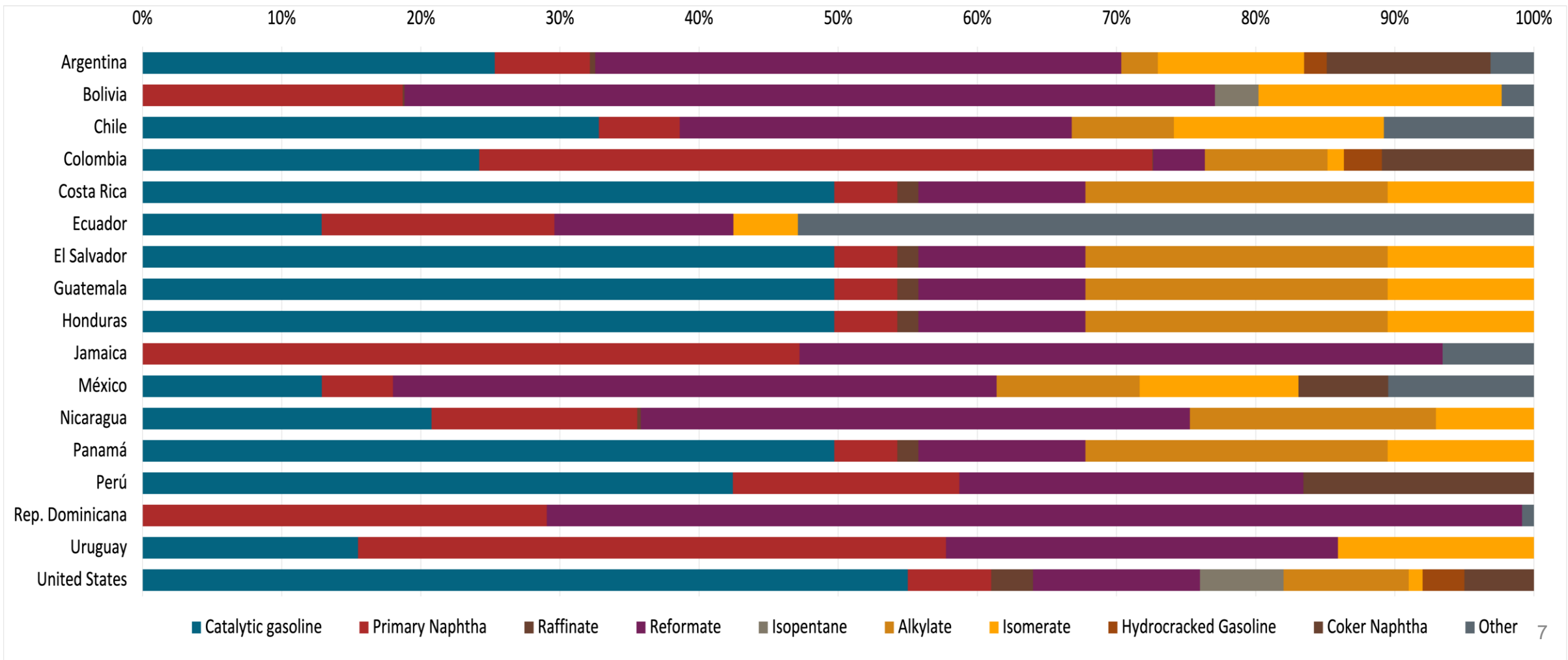
Gasoline Quality in El Salvador

Name	RTCA 75.01.19:19	RTCA 75.01.20:19	EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2021	2021	2017			
Applicability	Whole country	Whole country	All countries			
Selected Grade	Gasoline Regular	Gasoline Premium	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	5% v/v	5% v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	50% v/v	50% v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	30% v/v	30% v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	2,0% v/v	2,0% v/v	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 91	> 95	> 95	> 95	> 98	> 98
MON	-	-	> 85	> 88	> 85	> 88
AKI						
Sulfur Content	< 500 mg/kg	< 500 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	0,7% v/v	0,7% v/v	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	-	-	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	< 69 kPa	< 69 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)						
RVP 37.8°C (Transition)						
MTBE	-	-	-	-	-	-
Ethers 5 or more C Atoms	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

Source: RTCA

Gasoline Component Blending in Latin America

Gasoline is a blend of a base gasoline and other components. This blending is usually done at blending terminals as only 30% of the world's finished gasoline is distributed directly from refineries. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Latin America are:



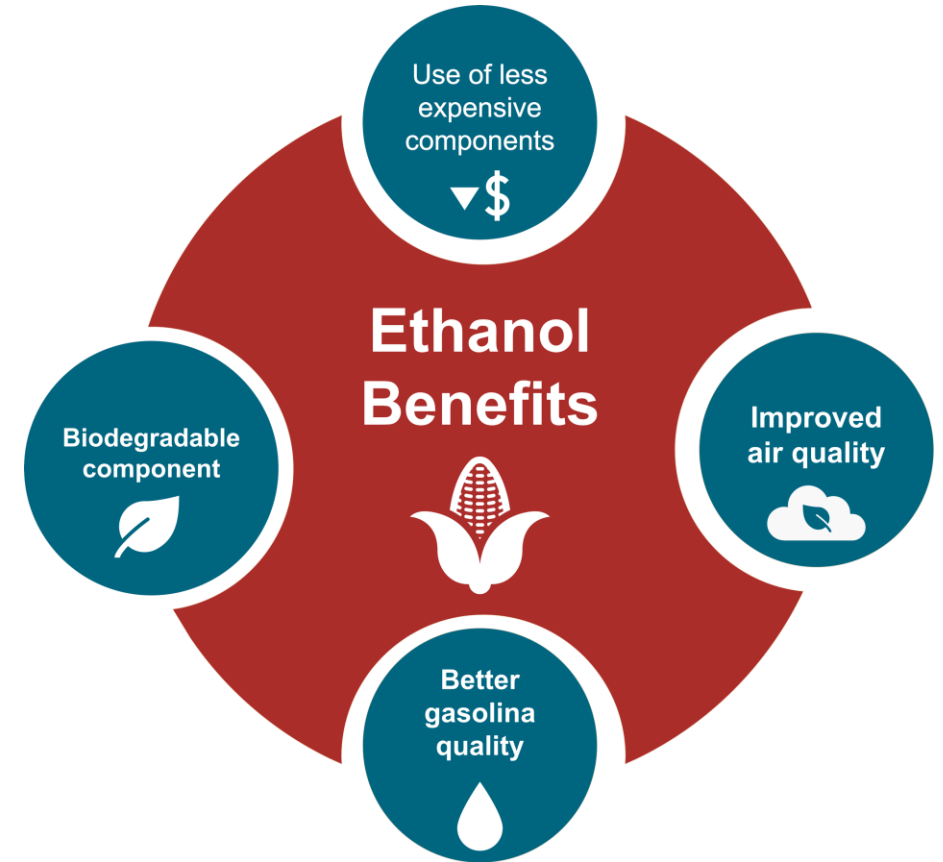
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

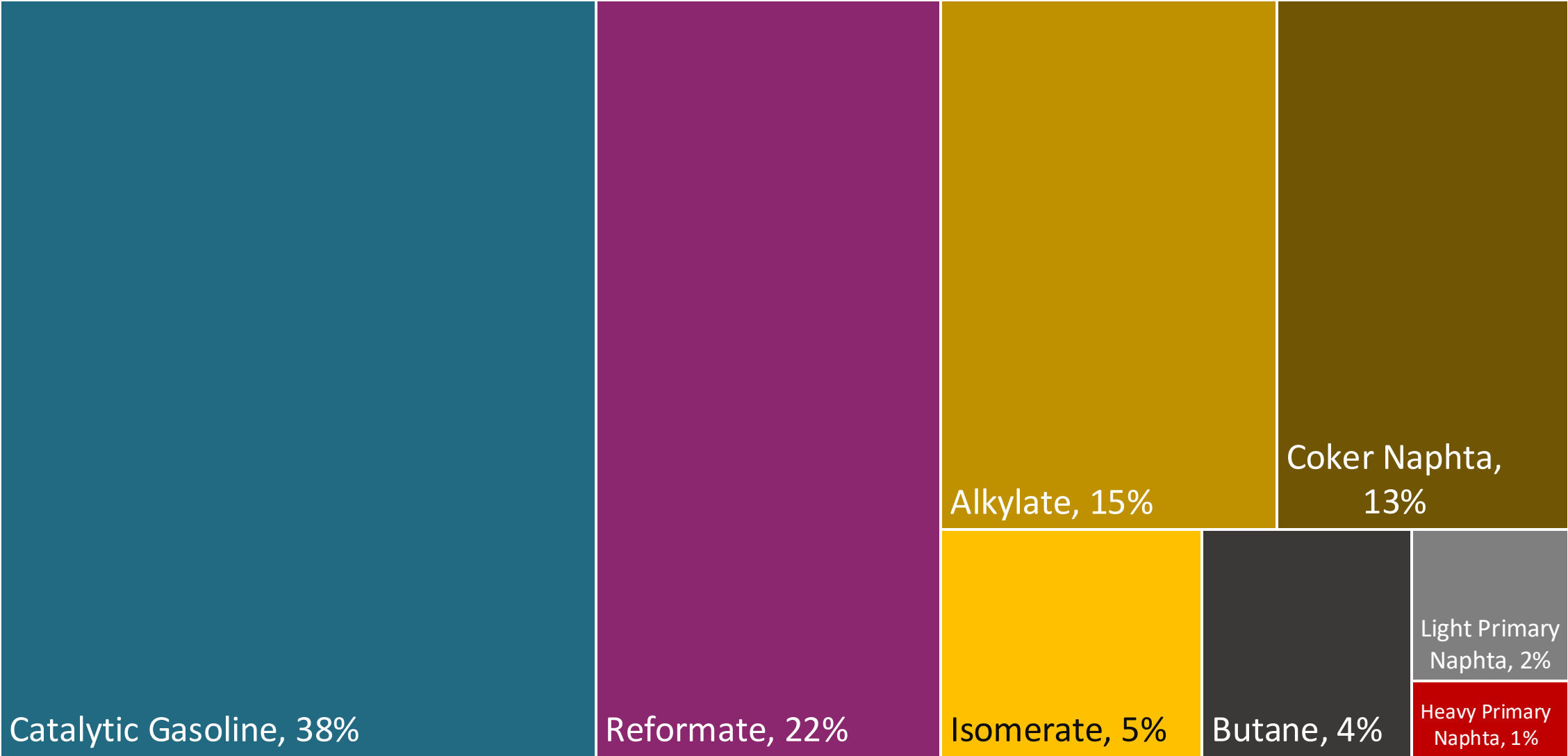
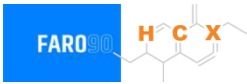
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average 2024 prices and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

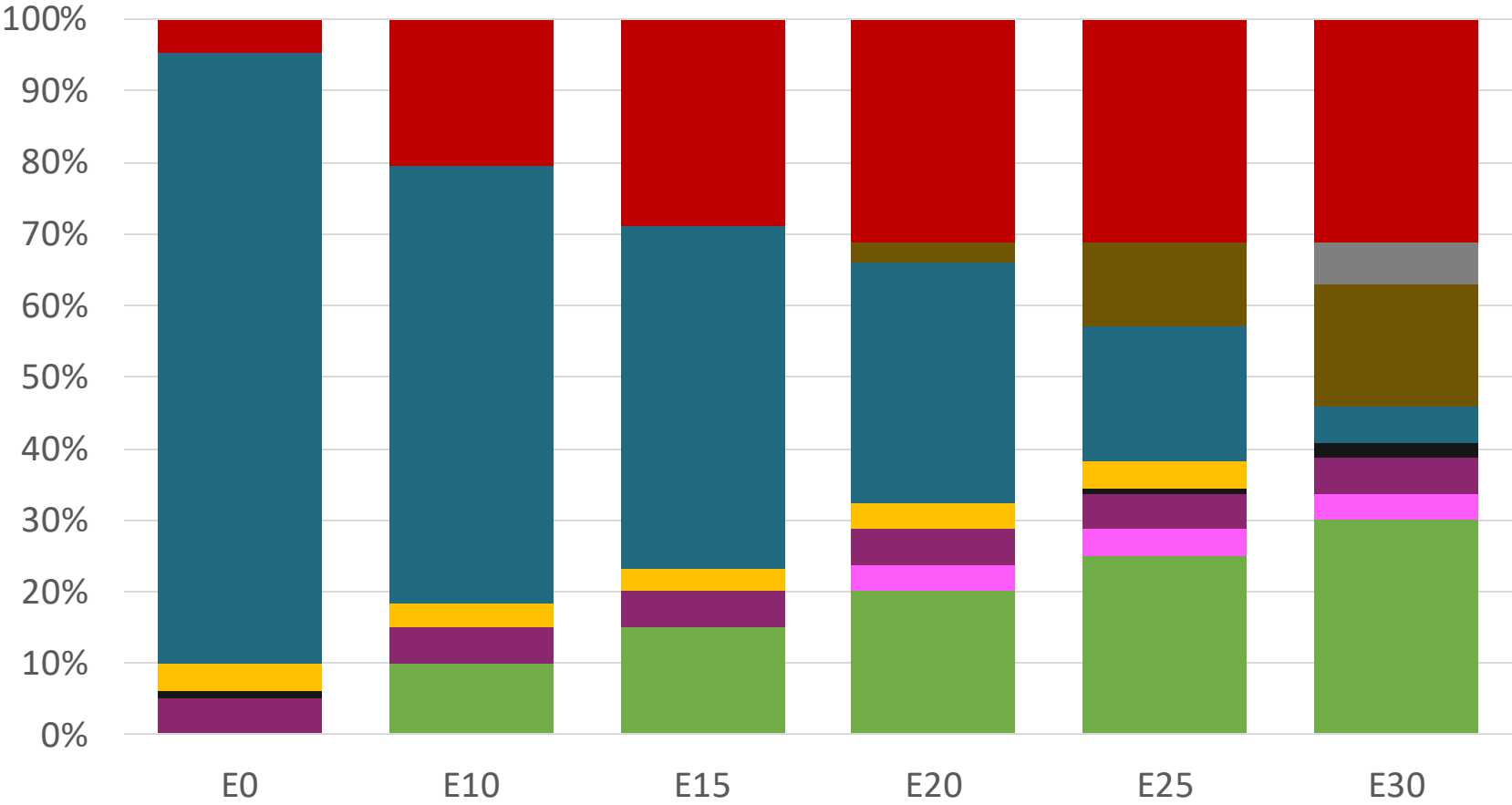
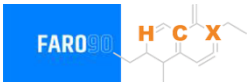


Blending Components- El Salvador



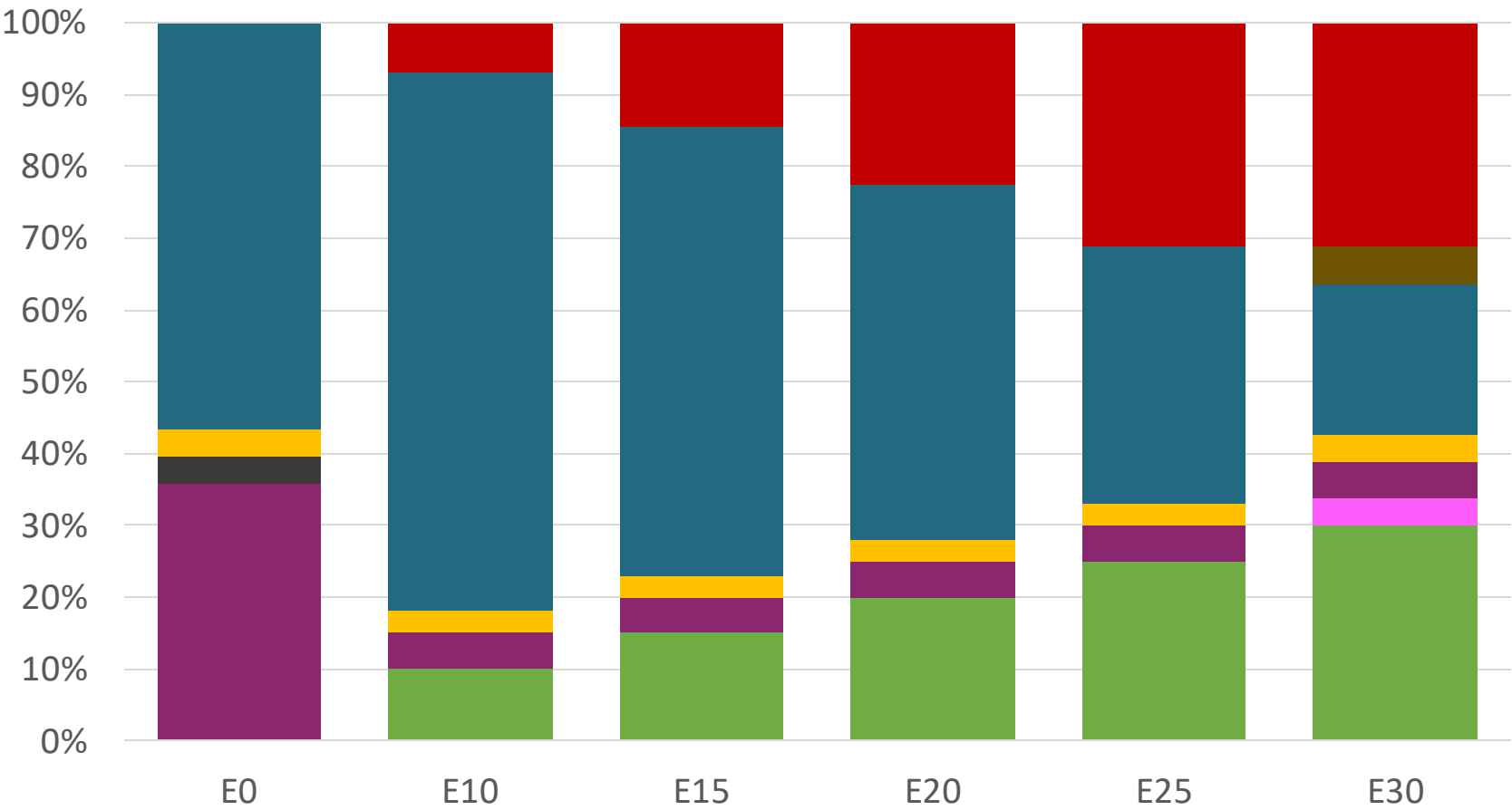
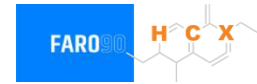
Source: Faro90

El Salvador – Regular – Constant Octane



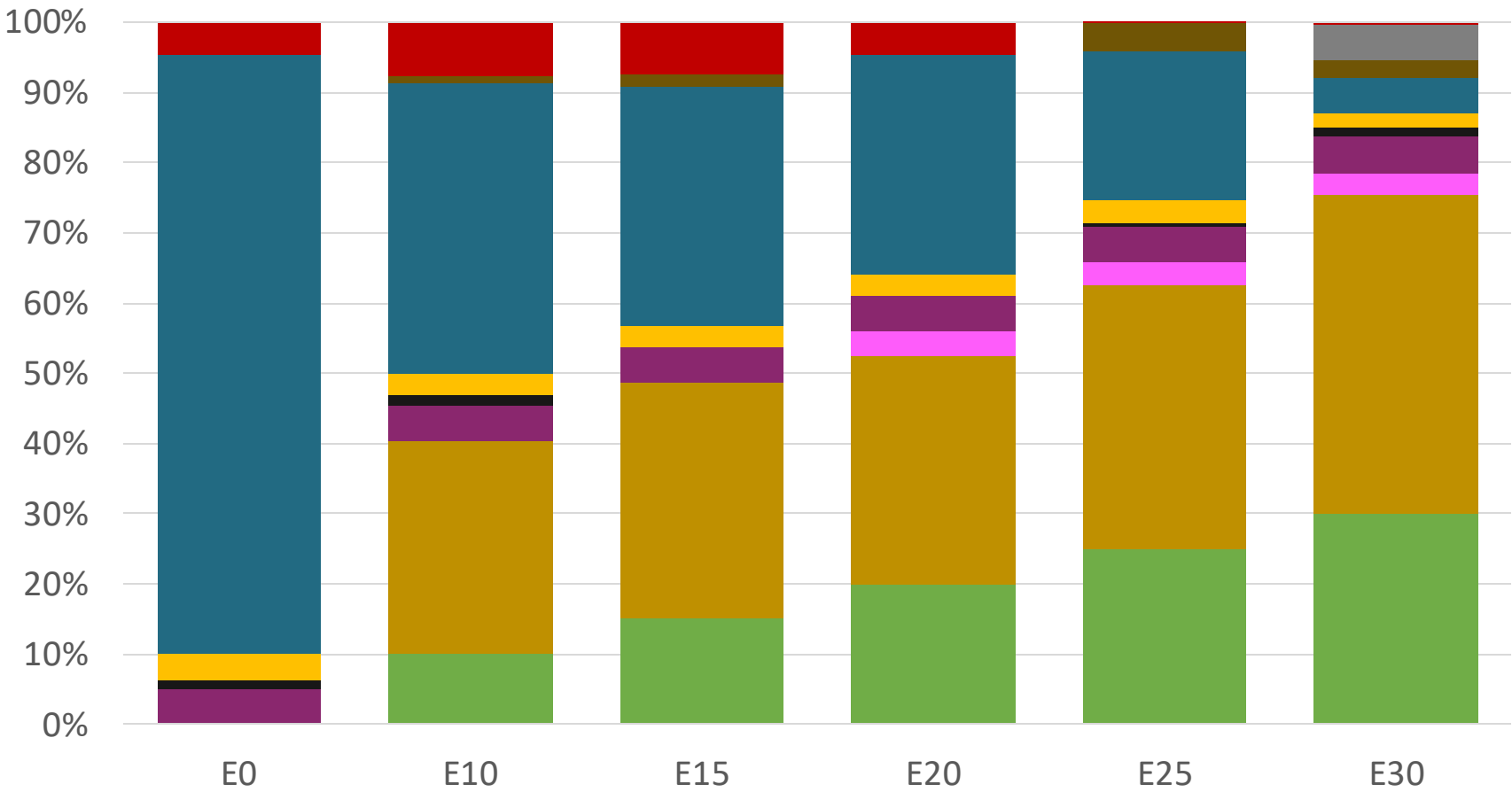
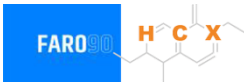
Octane (RON)	91.0	91.0	91.0	91.0	91.0	91.0
Price (USD/gal)	\$ 2.336	\$. 2.078	\$. 1.934	\$. 1.787	\$. 1.636	\$. 1.518

El Salvador – Premium – Constant Octane



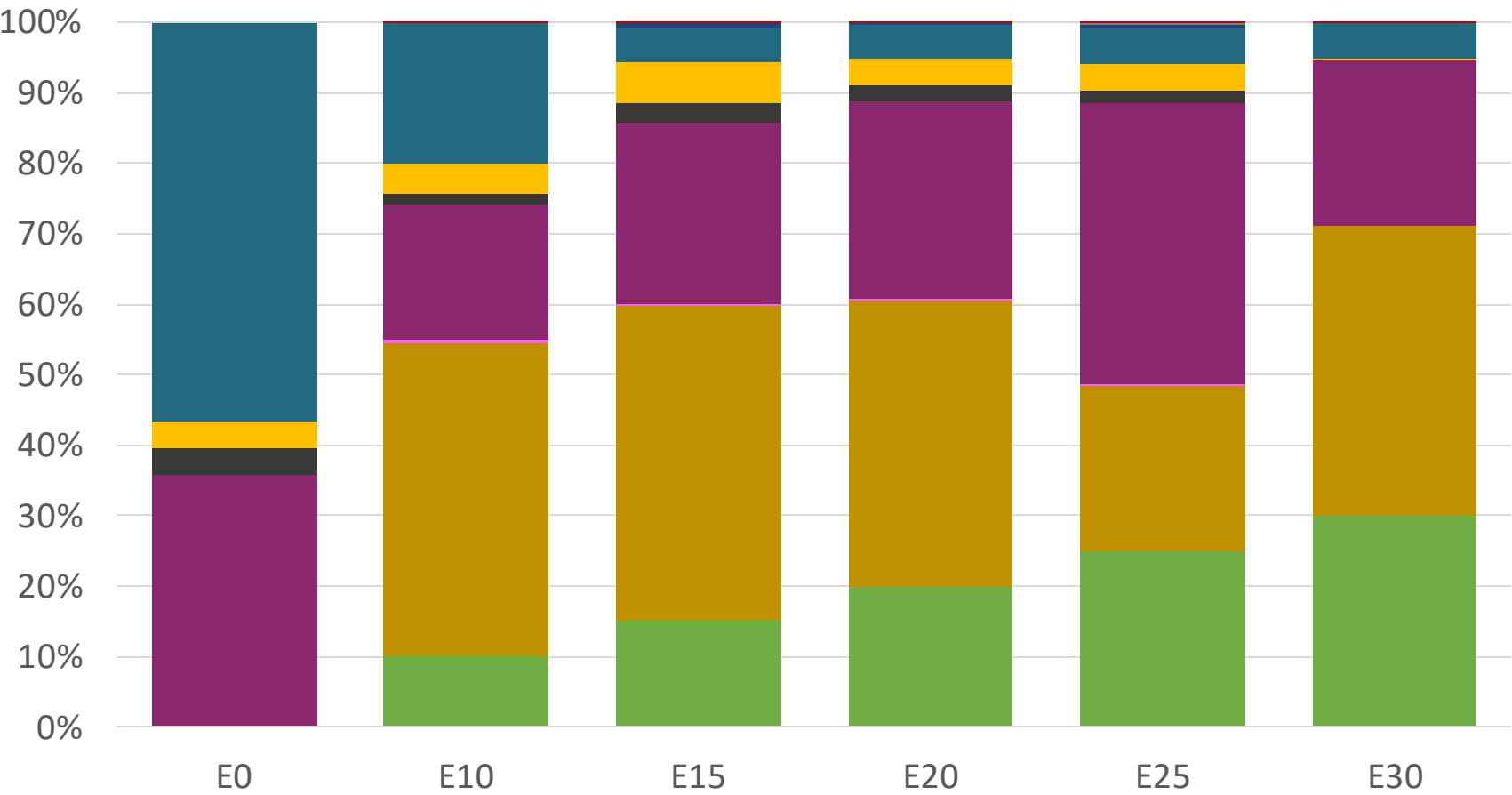
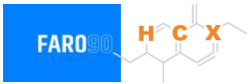
Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (USD/gal)	\$ 2.586	\$ 2.283	\$ 2.149	\$ 2.007	\$ 1.857	\$ 1.709

El Salvador – Regular – Increased Octane



Octane (RON)	91.0	94.6	96.6	99.1	101.2	102.4
Price (USD/gal)	\$ 2.336	\$ 2.336	\$ 2.336	\$ 2.336	\$ 2.336	\$ 2.336

El Salvador – Premium – Increased Octane



Octane (RON)	95.0	98.0	100.3	102.6	104.9	105.8
Price (USD/gal)	\$ 2.586	\$ 2.586	\$ 2.586	\$ 2.586	\$ 2.586	\$ 2.586

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

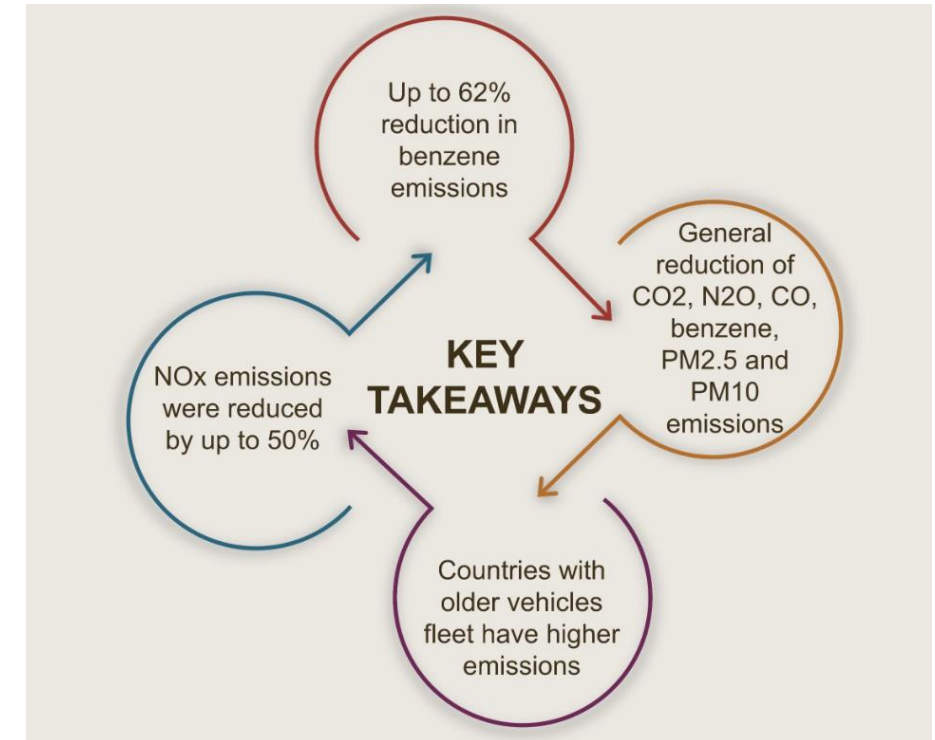
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

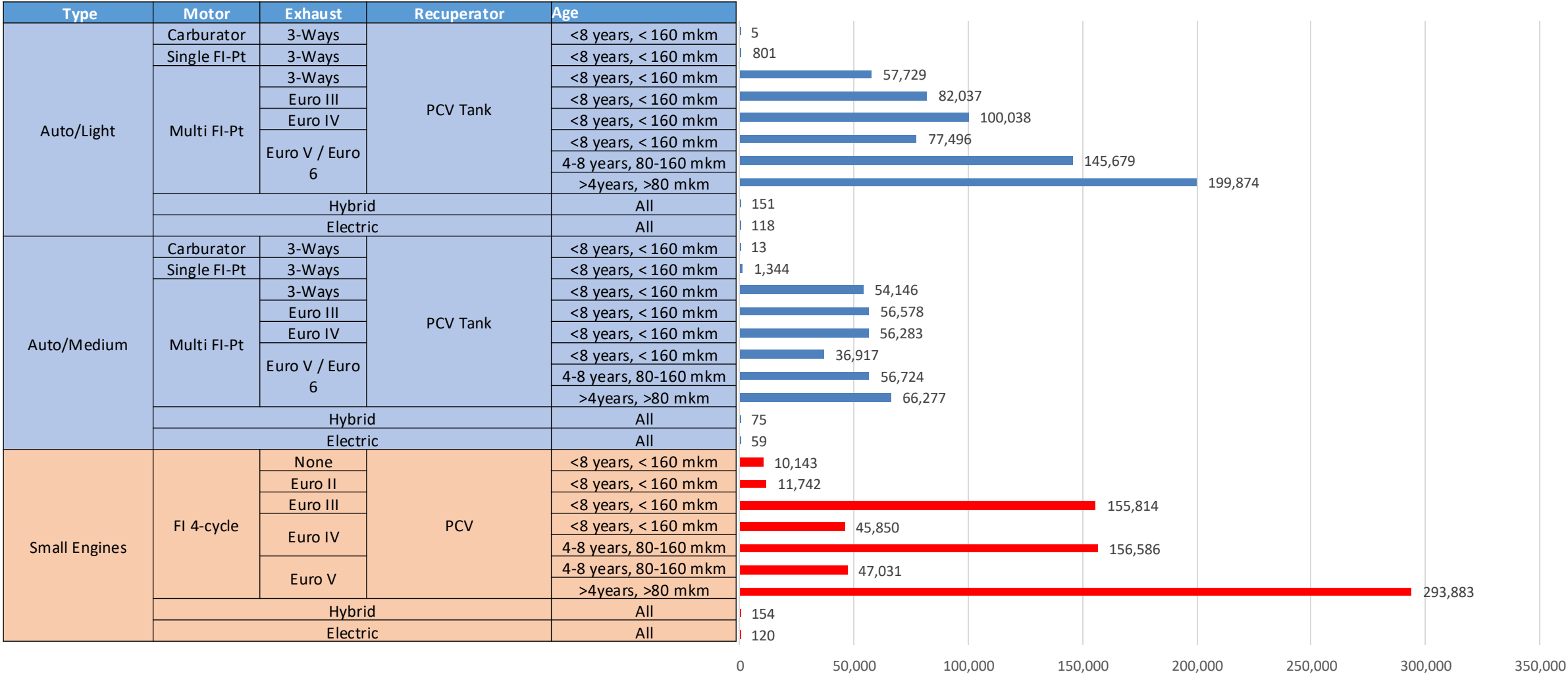
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



*Source: Bureau of transportation statistics.

Gasoline Vehicular Fleet in El Salvador



Vehicular Fleet: 1,712,991

Average Age: 9 years

Motorcycles: 33%

Source: Registro Publico, Faro 90

El Salvador - Vehicular Emissions



Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	159,839	149,803	139,768	131,984	124,352	118,546	111,473	-13%	-22%
CO2	6,127,680	5,970,111	5,821,296	5,704,257	5,646,342	5,621,675	5,518,052	-5%	-8%
VOC	27,009	25,783	24,556	23,717	22,949	22,381	21,542	-9%	-15%
NOx	11,653	8,740	8,157	7,688	7,251	6,768	6,258	-30%	-38%
SOx	73	68	62	57	53	48	43	-15%	-28%
NH3	2,340	2,317	2,294	2,305	2,331	2,349	2,364	-2%	0%
N2O	74	74	74	74	76	76	77	-1%	2%
CH4	6,335	6,335	6,335	6,430	6,569	6,689	6,803	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	132	124	116	110	104	100	93	-12%	-21%
Acetaldehyde	369	424	622	947	1,290	1,474	1,742	68%	249%
Formaldehyde	1,457	1,417	1,651	1,939	2,031	2,247	2,447	13%	39%
Benzene	392	377	357	352	349	333	323	-9%	-11%
PM2.5	396	352	307	267	227	184	139	-22%	-43%
PM10	1,033	917	802	696	591	480	364	-22%	-43%

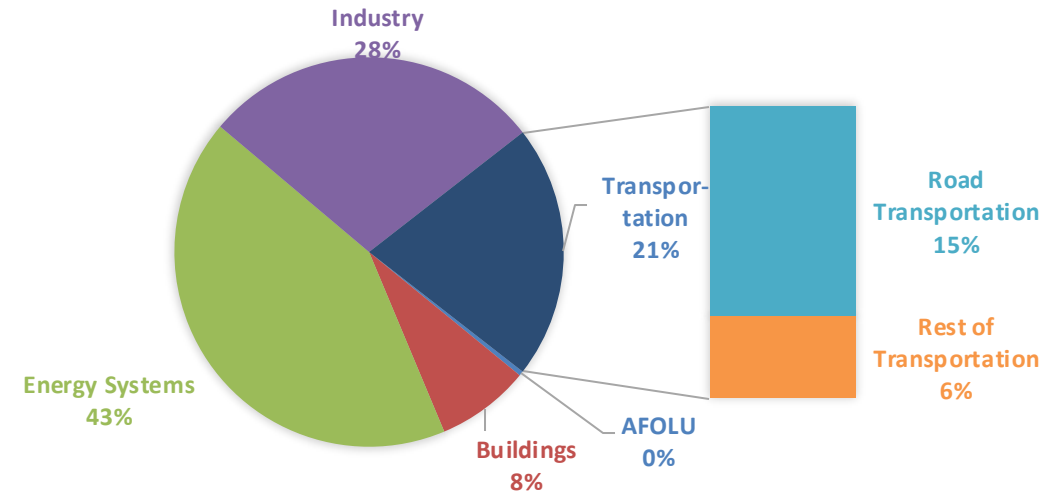
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

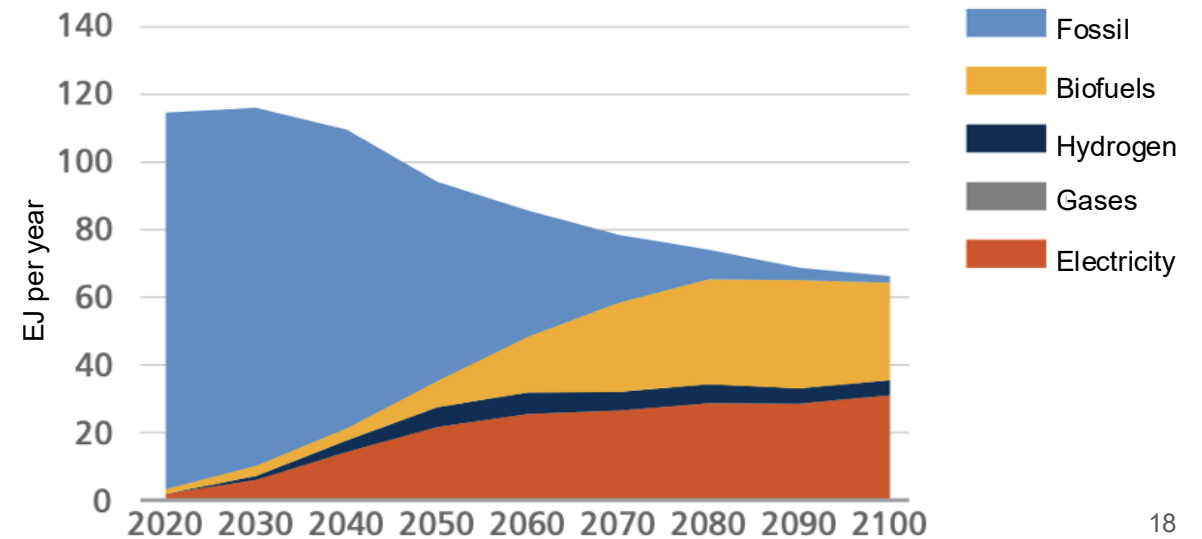
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

Current GHG Global Emissions Distribution

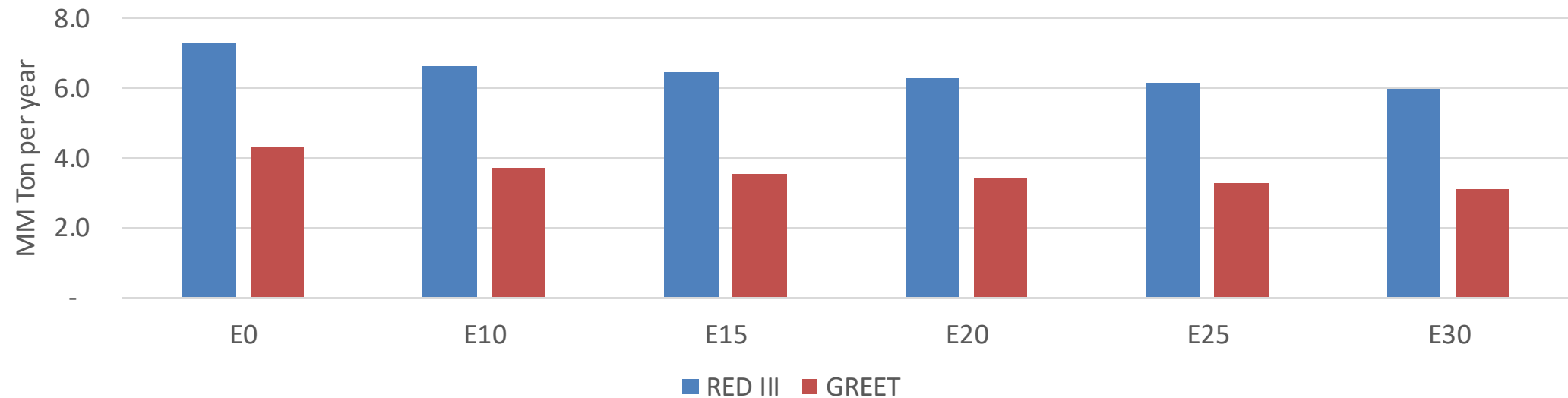


IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



El Salvador – Gasoline Vehicle Fleet GEI Emissions

Life Cycle GHG Emission - Gasoline Vehicles



Country Emissions 2024 (MMT/year)	14.8
2035 Target (MMT/year)	10.3
Delta	4.4

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	14.3%	17.9%	21.1%	24.1%	27.9%
RED II/III	0.0%	9.2%	11.7%	13.8%	15.9%	18.3%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	13.9%	17.5%	20.5%	23.5%	27.2%
RED II/III	0.0%	15.1%	19.3%	22.8%	26.2%	30.3%